

Sample Question Paper 1

Q1.

(a) Describe the concepts of User Interface (UI) and User Experience (UX), and distinguish between them by highlighting two key differences with suitable examples from a mobile application. **(2 Marks)**

(b) Critically evaluate the effectiveness of the Design Thinking process as a problem-solving methodology in UI/UX design projects, and support the evaluation with two real-world examples demonstrating its success or failure. **(5 Marks)**

Q2.

(a) A hospital website displaying lab reports uses very small font and dense text blocks, making it hard for patients to read. Organize typography by defining font size, spacing, and hierarchy. **(2 Marks)**

(b) Design a complete prototyping and usability testing plan for a telemedicine app for rural users with limited digital literacy. Include prototype fidelity rationale, five task scenarios, heuristic evaluation criteria, and a post-test iteration strategy. **(5 Marks)**

Q3. Attempt ANY 2 out of 3

(a) Apply the five-stage Design Thinking process to improve the checkout flow of an e-commerce site with a 65% cart abandonment rate, and describe the actions taken at each stage. **(4 Marks)**

(b) A university library portal is difficult to navigate, and students struggle to find books. Construct a survey by selecting any five questions to collect feedback on search and navigation issues. **(4 Marks)**

(c) In a UPI payment app, users are unsure whether the transaction is processing or failed because no clear feedback is shown during delays. Dissect feedback issues and explain how system status, progress indicators, and messages can improve user trust. **(4 Marks)**

Q4. Attempt ANY 2 out of 3

(a) A fitness center is launching a mobile app for its members. Construct a persona including any four elements (e.g., name, goals, frustrations, behavior). **(4 Marks)**

(b) A news website homepage contains multiple sections (politics, sports, entertainment) but lacks clear grouping and structure, making navigation confusing. Dissect content organization issues and suggest improvements. **(4 Marks)**

(c) You are designing a food ordering app in Figma. Apply interactive prototyping techniques to simulate the complete order placement flow, from menu browsing to order confirmation. **(4 Marks)**

Sample Question Paper 2

Q1.

(a) A startup is developing a new mobile application and wants to understand user needs before launch. List any four user research techniques and state what type of data each technique collects. **(2 Marks)**

(b) A smart city mobile app collects user feedback about public services (water, electricity, roads), but the development team is unsure whether to rely more on surveys or field observations. Evaluate both research methods and justify which method is more effective for identifying real user issues in this scenario (give any three points). **(5 Marks)**

Q2.

(a) You are designing a login screen in Figma. Your mentor asks you to quickly show how the button will look when clicked. Recall the prototyping feature you would use. **(2 Marks)**

(b) An e-commerce analytics dashboard used by business managers displays sales data, customer insights, and alerts on a single screen without proper grouping, hierarchy, or alignment. Managers struggle to quickly identify important trends and make decisions. Design and develop an improved dashboard layout, clearly explaining how you will apply grouping, alignment, visual hierarchy, and spacing to enhance clarity and usability. **(5 Marks)**

Q3. Attempt ANY 2 out of 3

(a) Analyze a task flow diagram for an e-commerce checkout process and identify inefficiencies affecting user completion. **(4 Marks)**

(b) A mobile game is frequently uninstalled within one day of download. Analyze user behavior and infer any four reasons for this behavior. **(4 Marks)**

(c) A hospital patient mobile app shows different layouts for appointment, reports, and billing sections. Analyze inconsistency issues and explain how design systems improve uniformity. **(4 Marks)**

Q4. Attempt ANY 2 out of 3

(a) An online pharmacy website uses bright and inconsistent colors across categories like medicines, health products, and offers, reducing readability. Investigate color imbalance and suggest improvements based on harmony and contrast. **(4 Marks)**

(b) A usability test found that users took 4 minutes to complete a 3-step checkout. Analyze the design bottlenecks and propose front-end improvements using HTML/CSS best practices. **(4 Marks)**

(c) A mobile banking app has frequent user errors during fund transfers. Apply heuristic evaluation to identify the usability violations and propose an improved interaction design. **(4 Marks)**

Sample Question Paper 3

Q1.

(a) Identify the five stages of the Design Thinking process and briefly state the goal associated with each stage. **(2 Marks)**

(b) Design a complete UX flow for an investment application targeting first-time investors with low financial literacy using the Design Thinking framework, and justify key UI and UX decisions made throughout the process. **(5 Marks)**

Q2.

(a) An e-commerce website for medicines uses random colors across product pages, reducing trust and clarity. Develop a consistent color palette including primary, secondary, and accent colors. **(2 Marks)**

(b) Evaluate the impact of skipping usability testing in a government healthcare portal. Analyze the consequences for user adoption, error rates, and accessibility compliance, and propose a minimal viable testing strategy under budget constraints. **(5 Marks)**

Q3. Attempt ANY 2 out of 3

(a) Apply UX feedback and error recovery principles to enhance a mobile banking app interaction where transaction errors occur without user-facing messages or recovery options. **(4 Marks)**

(b) A cab booking service wants to improve the experience for first-time users. Construct a user journey by listing any five key steps from app installation to ride completion. (explanation is required) **(4 Marks)**

(c) A hospital mobile application lacks predefined UI components, leading to inconsistent screens across modules. Analyze the problem and explain how atomic design can structure components effectively. **(4 Marks)**

Q4. Attempt ANY 2 out of 3

(a) A smart parking system app allows users to find and reserve parking spots, but users report difficulty in locating available slots. Construct a survey by selecting any five questions to understand issues in searching and booking parking spaces. **(4 Marks)**

Marks)

(b) You are converting a Figma product listing page into a responsive front-end. Apply CSS Grid to build a 3-column product card layout with a collapsible sidebar filter, describing the HTML structure and key CSS rules. **(4 Marks)**

(c) In usability testing, users cannot identify clickable elements. Apply appropriate design changes to improve affordance in your design. **(4 Marks)**

Sample Question Paper 4

Q1.

(a) A designer is creating an empathy map for users of a food delivery app. Identify the main sections of an empathy map and describe what is included in each section. **(2 Marks)**

(b) A smart education platform wants to improve the learning experience for remote students who face difficulty staying engaged. Design and construct a user journey map by organizing: 1. Any four stages of the user experience 2. One pain point and one opportunity at each stage. **(5 Marks)**

Q2.

(a) You designed a webpage layout in Figma, and now you need to convert it into a webpage. List the basic front-end technologies you will use. **(2 Marks)**

(b) A university web portal used by students and faculty uses multiple font styles randomly across pages, affecting readability and professionalism. Users find it difficult to distinguish headings, notices, and important information. Construct a typography system defining font hierarchy, font sizes, styles, and usage rules to improve readability and consistency. **(5 Marks)**

Q3. Attempt ANY 2 out of 3

(a) A startup launches a product without proper prototyping and user testing, leading to failure. Analyze the importance of prototyping and testing stages in the Design Thinking process and justify how they could have prevented the failure. **(4 Marks)**

(b) Users of a digital wallet app feel confused while adding money to their account. Identify and explain any four usability problems faced by users in this process. **(4 Marks)**

(c) A university website displays notices, events, and announcements without proper spacing, making content difficult to scan. Break down spacing issues and explain how spacing improves readability and grouping. **(4 Marks)**

Q4. Attempt ANY 2 out of 3

(a) A startup is designing a new mobile app and wants to maintain consistency across all screens. Construct an example of Atomic Design structure by organizing components into atoms, molecules, and organisms for a login screen. **(4 Marks)**

(b) Apply usability testing methodology to design a test plan for a library management system. Include test objectives, participant criteria, task scenarios, success metrics, and a reporting format. **(4 Marks)**

(c) During usability testing, users repeatedly make errors while filling a form. Analyze the design and suggest improvements. **(4 Marks)**

Sample Question Paper 5

Q1.

(a) Identify any four commonly used UI/UX design tools and describe the primary purpose of each. **(2 Marks)**

(b) Design a user-centered ATM interface tailored for elderly and differently-abled users, incorporating accessibility features, usability principles, and inclusive design considerations with appropriate justification. **(5 Marks)**

Q2.

(a) A hospital software development team repeatedly designs buttons, cards, and forms separately for each module. Apply atomic design to organize reusable UI components. **(2 Marks)**

(b) Create a front-end implementation plan for converting a 5-screen Figma prototype of a job portal into a responsive React application. Detail component hierarchy, CSS design token usage, responsive breakpoints, and how the Figma handoff informed each decision. **(5 Marks)**

Q3. Attempt ANY 2 out of 3

(a) A hospital management system is technically efficient but doctors find it stressful and time-consuming to use. Evaluate the role of UX in system-level architecture and recommend changes to improve overall system efficiency and user satisfaction. **(4 Marks)**

(b) A travel booking website notices users leave the app at the payment stage. Analyze the situation and infer any five reasons for user drop-off. **(4 Marks)**

(c) A hospital appointment booking mobile app is facing issues where patients are confused while selecting doctors, dates, and time slots due to poor layout and unclear flow. Analyze the existing usability issues and draw a Mid-Fidelity Wireframe showing improved screen layout for doctor selection, date/time selection, and booking confirmation. Clearly label all sections and briefly justify your design choices. **(4 Marks)**

Q4. Attempt ANY 2 out of 3

(a) A supermarket app allows users to scan items and pay without billing counters.

Develop a task flow showing the main steps from scanning items to payment confirmation. **(4 Marks)**

(b) A design team skips wireframing and directly creates high-fidelity designs (Prototype), causing confusion and rework. Analyze the issues and explain the importance of wireframing. **(4 Marks)**

(c) A UX team created a paper prototype for a smart home app. Analyze what user feedback can be collected at this fidelity level and explain its limitations compared to a high-fidelity Figma prototype. **(4 Marks)**